

IRIS DATASET - 4D

3 LABELS

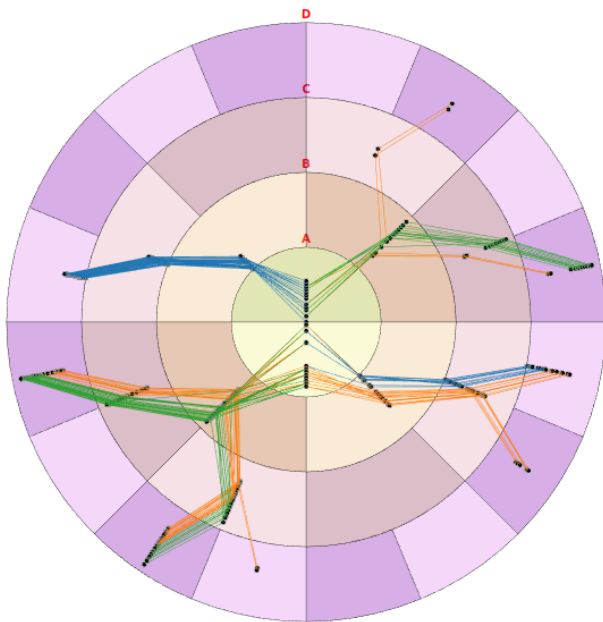
A = SepalWidthCm

B = SepalWidthCm_PetalWidthCm

C = SepalWidthCm_PetalWidthCm_SepalLengthCm

D = SepalWidthCm_PetalWidthCm_SepalLengthCm_PetalLengthCm

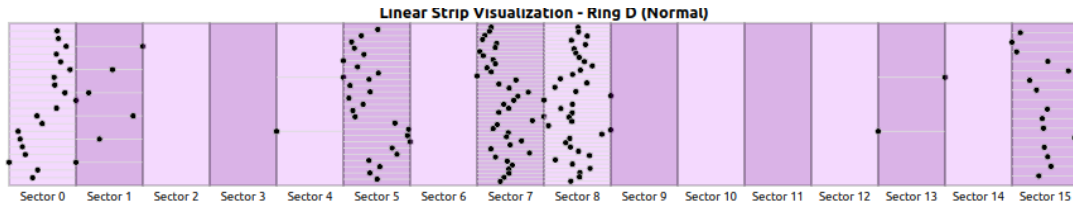
■ Iris-setosa
■ Iris-versicolor
■ Iris-virginica



NORMAL VIEW

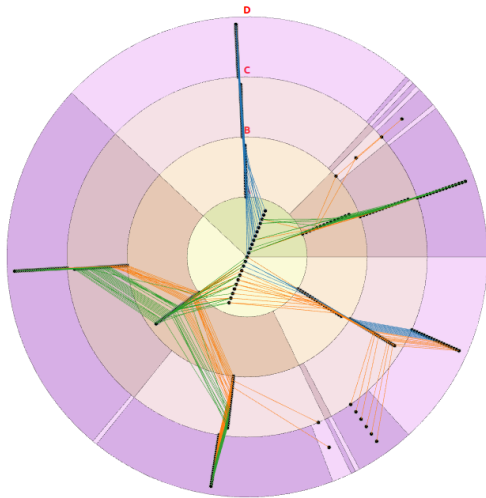
Observation:

- Cluttering of one type of label together
- Usage of only certain quadrants



Observation:

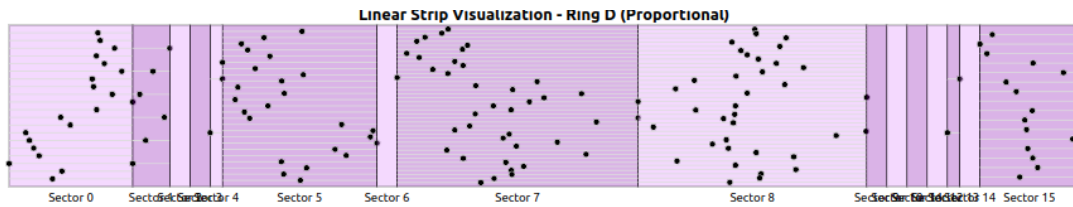
- Distribution of points in terms of their value



PROPORTIONAL VIEW

Observation:

- Shows the usage of quadrants by making the quadrant size proportional to number of points.



Observation:

- The distribution of points in a sector which has area proportional to the number of points; gives us an idea of the variability of point value in the sector of the last ring.

DIABETES DATASET - 7D
2 LABELS

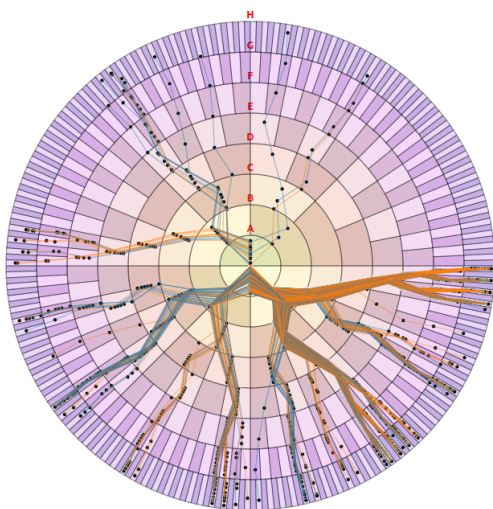
A = DiabetesPedigreeFunction

B = DiabetesPedigreeFunction_Pregnancies

C = DiabetesPedigreeFunction_Pregnancies_BMI

D = DiabetesPedigreeFunction_Pregnancies_BMI_Age

E = DiabetesPedigreeFunction_Pregnancies_BMI_Age_SkinThickness



NORMAL VIEW:

Observation:

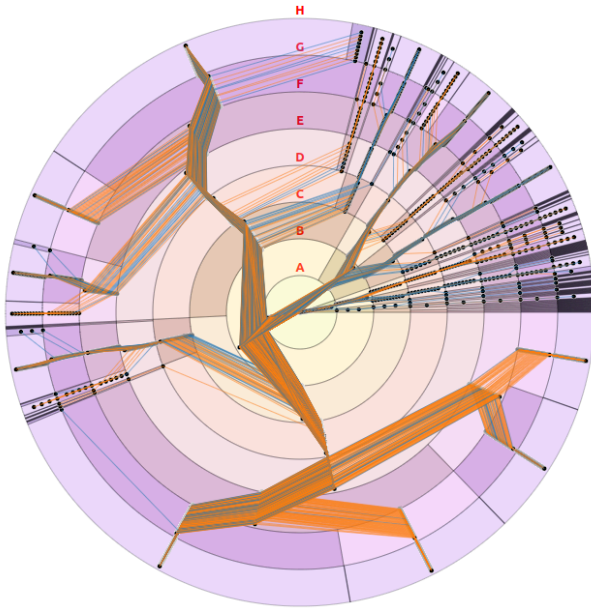
- Many empty sectors (uniform area of sectors)
- Shows that the many points of the feature A (Pedigree function) have a negative coordinate.

Drawback:

- Difficult to make the linear strip due to limited area.

Solution:

1. Heat Map strip to show the number of point range.
2. Omitting the empty sectors completely and relying on sector number solely.
3. Simultaneously making the empty sectors in the main viz of the same colors (say red)



Observation:

- Usage of negative quadrant in ring A is over 90%

Drawback:

- Deflection due to sector size could be misleading
- Transparency of edge is required to avoid confusion and wrong interpretation

Solution:

1. Setting a transparency for the edges.
2. Need to find solution for solving mis-interpretation of the deflection or change in route of points when they move from one ring (subpace) to another.

RANDOMLY GENERATED 0/1 DATASETS

10 DIMENSIONS - 5 LABELS - 1000 POINTS

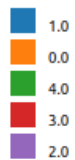
A = Feature_10

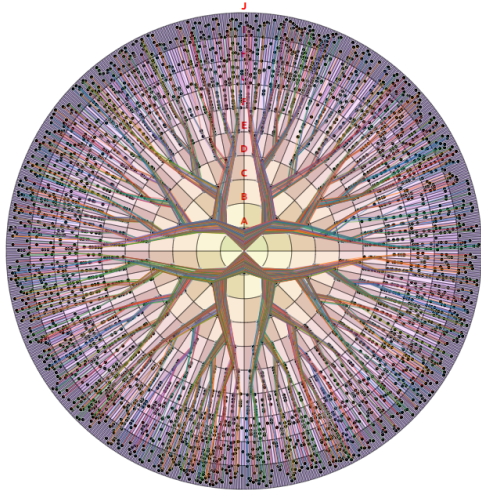
B = Feature_10_Feature_3

C = Feature_10_Feature_3_Feature_4

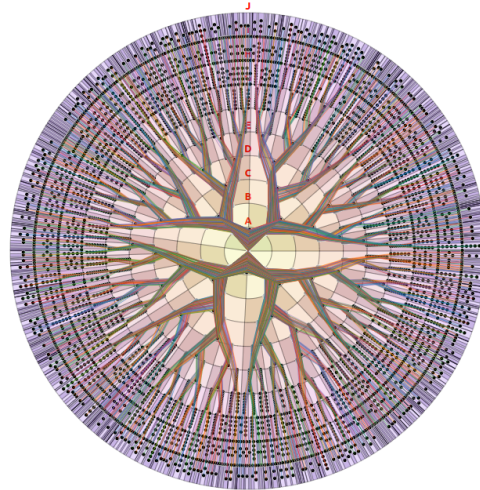
D = Feature_10_Feature_3_Feature_4_Feature_9

E = Feature_10_Feature_3_Feature_4_Feature_9_Feature_1





Normal View



Proportional View

10 DIMENSIONS - 2 LABELS - 512 POINTS

A = Feature_9

B = Feature_9_Feature_10

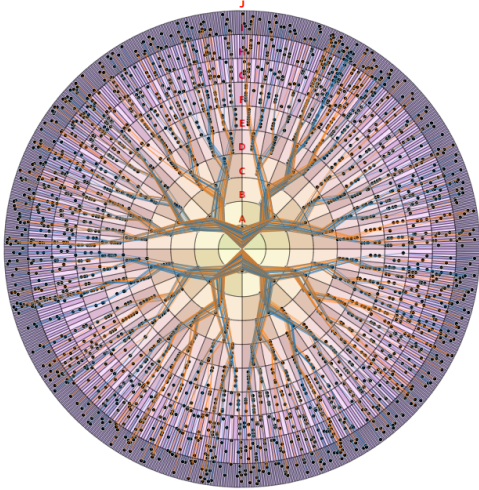
C = Feature_9_Feature_10_Feature_1

D = Feature_9_Feature_10_Feature_1_Feature_4

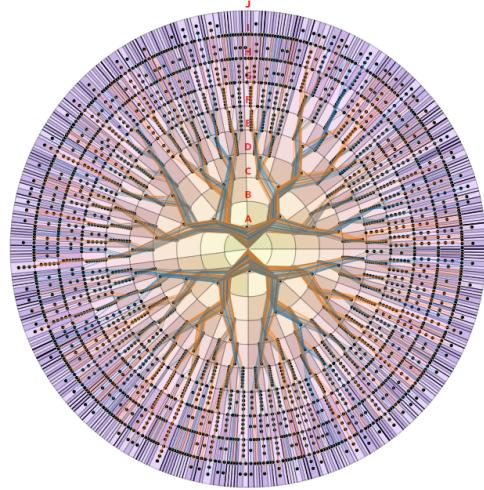
E = Feature_9_Feature_10_Feature_1_Feature_4_Feature_8

1.0

0.0



Normal View



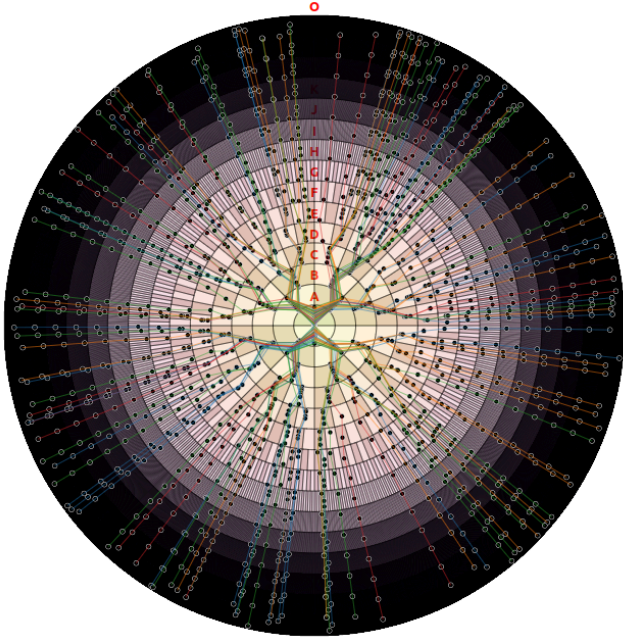
Proportional View

Observation:

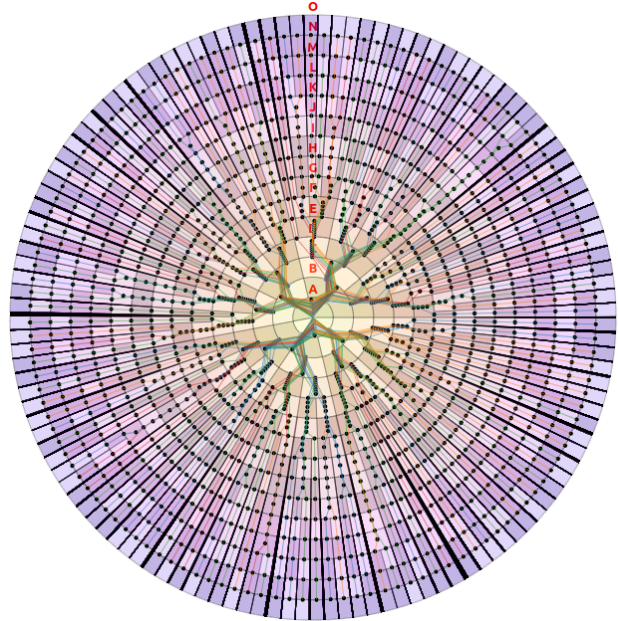
- Doesn't seem like much of a difference but the proportional view gives a clearer view of the distribution.
- If the sector is observed, it gives an idea about the quadrant usage (number of points lying in the sector)

Here is the data to validate the above observation

The following is a 15 Dimensional 0/1 Dataset randomly generated with 4 labels in it and a total of 128 data points



Normal View



Proportional View

Observation:

1. The problem of 360 degree being divided into 2^d sectors makes the area of the sector very negligible.
 - a. Suppose we have 15 dimensional data.
 - b. Number of sectors = $2^{15} = 32768$ sectors
 - c. Angle allocated to each sector in uniform distribution = 0.01 degree which is negligible
2. The proportional view helps in the following way:
 - a. by using pigeonhole principle at least $(32768 - 128 = 32640)$ sectors which will be empty will be discarded
 - b. This saves us $32640 * 0.01 = 326.4$ degrees if the empty sectors are removed from the visualization.
3. This also helps us
 - a. visualize higher dimensional data with reasonable number of points (limitation : number of labels and number of points -> the approximate number of these is to be figured out)